

The Math Behind the Melody

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Introduction

Music. It is typically defined as expression of sound placed together in a space with notions of harmony, melody and rhythm. However these qualities of music only go as far as to describe how beautiful it is. For some, the flow of music is the method of which time can pass by quickly in a pleasant package. For others it is their passion, the craft that one spends a considerable amount of time improving on. But all in all, there is no doubt that music ensnares everybody and that words only go so far to articulate the feeling.

From this point however one may be inclined to ask, what is the secret behind music? What exactly makes each cog in this system turn? Historically, the pioneers behind music such as Mozart and Bach have walked the line between utilizing and building upon music theory, and going of what they had felt was right. This could not be further applied to a composer such as Beethoven, who upon going deaf was still able to compose some of the world's greatest musical compositions. From these stories most may overlook that behind each harmony, each melody, and each rhythm lies a very elaborate corresponding system of mathematics that had been made for it. In other words, this fascinating art is really just an expression of mathematics. This paper aims to cover the art of music in a new perspective, one that can break it down to numbers and to analyze the aforementioned ideas of harmony, melody and rhythm as it pertains to the field of mathematics. Hopefully, the average mathematician might pick up a thing or two from music theory and vice versa.



Pitch

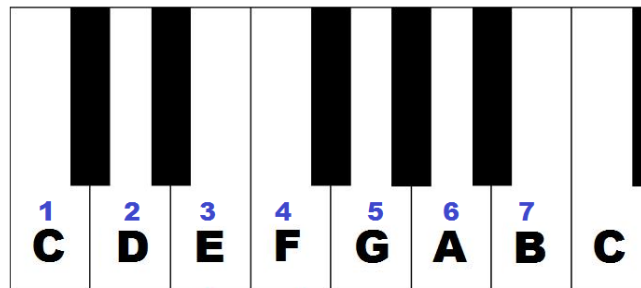
First, it is important to understand that music is just a series of sounds stitched together. Often, the more melodic sounds are made through manipulating the frequency of sound waves. One such example commonly used in music are strings. When tuning a string instrument such as the guitar, musicians would twist a peg located at the head of the instrument with which one can twist to their convenience and to harness their preferred tone. A combination of plucking the string and overhearing the sound it produces is how one can successfully tune a string to a specific frequency (in Hz)[1][2]. In music, certain notes are created at specific frequencies. Pitch is the distinction between which notes are "high" and which notes are "low". Understanding this is a great place to start when cultivating knowledge on music as a whole.

The following note on the piano is denoted as Middle C:



for which is often used as a center point in music theory. Middle C is obtained when a device or instrument is tuned to a frequency of 256 Hz (however in some instances we tune it to 261.625565)[2]. Musicians will typically tune other notes relative to this middle C. However, how is this done? Take the C Major Scale for example. Here, the C Major Scale is the most simple scale among all others in music theory as it only utilizes the white notes on a keyboard.

C Major Scale



In this diagram, C is the starting note of the scale, and what follows are the letters D, E, F, G, A, B and a higher pitched C (note that after G, by convention the letter scheme repeats back to A). It is important to mention that as one follows the scale from left to right, the frequency of each note increases and thus are perceived to have a higher pitch. The opposite occurs when following the scale from right to left .

Pythagoras had pioneered the notion of mathematical ratios between the sound of each note.[1] This relationship has been called an interval. It just so happens to be that the notes that work well together in music are tied with a specific ratio of their frequencies. In the given case of the C Major Scale:

C D E F G A B C

the ratios are given as:

$$\frac{9}{8} \quad \frac{10}{9} \quad \frac{16}{15} \quad \frac{9}{8} \quad \frac{10}{9} \quad \frac{9}{8} \quad \frac{16}{15}$$

[1][2][3]

Here, the ratio in frequency between the notes C and D are $\frac{9}{8}$. So if Middle C is represented as 256 Hz, the D note that follows this Middle C would have the frequency:

$$256 \times \frac{9}{8} = 288 \text{ Hz}$$

Tuning from C to E would follow as:

$$256 \times \frac{9}{8} \times \frac{10}{9} = 320 \text{ Hz}$$

and so on and so forth.

The following is an interesting application of this:

- Why do musicians specifically take the letters A to G and not more or less? This is often one of the first questions new musicians ask. To the casual musician and listener, one can hear that after seven straight notes of the scale, the eighth note would sound like the first note just pitched up, or up an octave in music theory. However mathematically this also checks out. The following is an A Major Scale:

A B C[#] D[#] E F[#] G[#] A

Note here, that similar to the C Major Scale, the first note of the scale (being A) is also the last note. To start, it is given that the ratios here follow the same pattern from C Major (note that C[#] is pronounced "C Sharp" which indicates the right black key above C and the same follows for the other black keys) :

$$\frac{9}{8} \quad \frac{10}{9} \quad \frac{16}{15} \quad \frac{9}{8} \quad \frac{10}{9} \quad \frac{9}{8} \quad \frac{16}{15}$$

Suppose that someone is to apply all the necessary operations to tune A to the next A. From Middle C, one can compute that A has a frequency of 440 Hz. Then by applying the same ratios:

$$= 440 \times \left(\frac{9}{8} \times \frac{10}{9} \times \frac{16}{15} \times \frac{9}{8} \times \frac{10}{9} \times \frac{9}{8} \times \frac{16}{15} \right) \tag{0.1}$$

$$= 440 \times (2) \tag{0.2}$$

$$= 880 \text{ Hz} \tag{0.3}$$

to which the higher pitched A is twice the frequency of the first A. Thus, as these ratios apply for any major scale in music, it will always be the case that the following octave is simply twice the frequency as the last.[3]

Rhythm

Perhaps one of the most captivating aspects of music is the groove or beat. It is easy to tap one's foot to the beat of a song as it is almost as easily recognized as the melody of a famous song. However is there a way to describe this sensation in terms of mathematics?

The quarter note is one of the most, if not the most recognizable symbol in music. It is symbolized as ♩ and is a basic unit of rhythm in musical pieces. It is important to remark that a quarter note is $\frac{1}{4}$ of a whole note. [2]

This basic unit in music is tampered with in order to "spice up" a piece by either holding down the note for a longer sound, or cut short for a fast, quick sound. Thus this quarter note could be manipulated in such a way that the following notes could be created:

♩							
♪				♪			
♪.						♪	
♪	♪	♪	♪	♪	♪	♪	♪
♪♪	♪♪	♪♪	♪♪	♪♪	♪♪	♪♪	♪♪
♪	♪	♪	♪	♪	♪	♪	♪

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where notes above ♩ dictate a longer note and ones below denote a shorter quicker sound. Each note in this chart are the following:

- Whole Note: ♩, which consists of four quarter notes and is $\frac{4}{4}$ of a whole note.
- Half Note: ♪, which consists of two quarter notes and is $\frac{1}{2}$ of a whole note.
- Quarter Note: ♩, which is $\frac{1}{4}$ of a whole note.
- Eighth Note: ♪, of which two eighth notes make up one quarter note and is $\frac{1}{8}$ of a whole note.
- Sixteenth Note: ♪, of which two sixteenth notes make up one eighth note and is $\frac{1}{16}$ of a whole note (this particular note is not within the chart above).
- ...

which can go on for so on and so forth. One can see that each unit divides the length of a whole note in powers of 2.[4] With this, it can be concluded that a beat of a song could be subdivided into as many parts and that a whole note can be divided into only so many parts that can be humanly performed or comprehended.

Composition

It is no surprise today that a song can be modified by any musician in order to produce a remix, a cover, or an entirely new song. One method of distinguishing an artist's aforementioned cover or remix is to modify the pitch of a song. Perhaps a vocalist is capable of singing incredibly high notes and so the band could perform songs at a higher pitch to accommodate this change. A heavy metal rock band could choose to

perform a song at a much lower pitch so the heavy guitars can really hit the hard riffs. This process is known in music theory as transposition.

Transposition, using the definitions built up from earlier, is simply the act of moving every note in the song by a specific interval.



As one can observe from this diagram, the overall "shape" of the melody does not change, however it is upon performing each scale separately that they do sound different. Each note in a line will have been shifted up or down in the same fashion as the rest to fit into another major scale. In mathematics, transposition fits perfectly well under group theory.

Refer back to the C Major Scale. Suppose the black keys are included, then there would be twelve unique notes:

$$\{C, C^\#, D, D^\#, E, F, F^\#, G, G^\#, A, A^\#, B\}$$

In music theory, a semitone is the distance between notes in this particular group. For example, C neighbors $C^\#$ and so they are one semitone apart. The note G is seven semitones above C and is two semitones below A. Since music is the manipulation of these twelve notes, it is found to be isomorphic to the group $\mathbb{Z}_{12}$ in which transforming this group preserves its properties such as order. T_n is referred to be the transposition transformation of a grouping of notes where n is of $\mathbb{Z} \bmod 12$ (as there are 12 unique notes). As an example, take the series of notes C, D, E in C Major where they create the harmony "do re mi":

$$\{C, D, E\}$$

we can thus we can represent the notes in their semitone position relative to the original note C:

$$\{0, 2, 4\}$$

where D and E are the second and forth semitones above C. Assume that a pianist intends to play the famous song in the key of A. Then, since A is nine semitones above C, $n = 9$ and we can increase each note in the melody as follows:

$$T_9 = \{0 + 9, 2 + 9, 4 + 9\} \tag{0.4}$$

$$= \{9, 11, 13\} \tag{0.5}$$

$$= \{9, 11, 1\} \tag{0.6}$$

to which C, D and E are then transposed to the notes A, B and $C^\sharp$ with a transposition of nine semitones above C. By method of comparing the ratio of frequencies or by analyzing the differences in semitones between the new three notes, it is shown that the semitones between the notes is preserved. Thus T_9 and T_n , in general case, is isomorphic.[2][4]

Conclusion

The essence of sound and nature completely collide in the world of music. It is shown that music is not only fundamentally based in a system of numbers, but is applied in a vast array of music theory as the world knows it today. It can be seen dipping its head in science through frequencies, and re-contextualized through group theory and simple algebraic work through fractions. All of this and the ever so evolving mold of music with the rise of technology and education will surface more interesting mathematical blends in the near future.



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